STEP 1
Order Entry
ERP sales order

STEP 2
Credit & Release
ERP order status

STEP 3
Pick & Pack
WMS

STEP 4
Carrier Tender
TMS

STEP 5
Pickup & Transit
Carrier EDI 214 / visibility

OUTCOME (Y)
Delivered On Time?
Proof of delivery

▲ Queue: awaiting credit/release | ▲ Queue: awaiting wave/pick | ▲ Queue: awaiting tender | ▲ Queue: awaiting pickup | ▲ In transit

Decision window: risk score must be available here to expedite, re-sequence, or re-tender | **Limited options** | **Too late to act**

STEP 1 — Order Entry

DECISION

Can we commit to the requested date?

QUESTIONS

- Is the requested date feasible against lead time?
- Is the item available to promise?

VARIABLES

- ◐ Requested vs. promised date gap
  - ↳ *ERP order header*
- ◐ Quoted lead time
  - ↳ *ERP item master*
- ◐ Available-to-promise qty
  - ↳ *ERP inventory and ATP*
- ✓ Discount offered
  - `Discount_offered`
  - ↳ *ERP pricing*
- ✓ Product importance
  - `Product_importance`
  - ↳ *Product master*
- ✓ Product cost
  - `Cost_of_the_Product`
  - ↳ *ERP item cost*

Primary system: ERP order header and ATP

STEP 2 — Credit & Release

DECISION

Release the order now or hold it?

QUESTIONS

- How long has the order been waiting?
- How much slack remains to the promise date?

VARIABLES

- ◐ Order-to-release hours
  - ↳ *ERP status log*
- ◐ Slack days to promise date
  - ↳ *ERP order header*
- ◐ Credit hold flag and duration
  - ↳ *ERP credit module*
- ✗ Reason code for hold
  - ↳ *Not recorded; email or notes*

Primary system: ERP status change log

STEP 3 — Pick & Pack

DECISION

Prioritize this order in the wave?

QUESTIONS

- Is the warehouse running behind?
- Is the item difficult to handle or ship?

VARIABLES

- ◐ Pick backlog at release
  - ↳ *WMS task queue*
- ✗ Labor utilization by shift
  - ↳ *Shift labor spreadsheet*
- ◐ Pick and pack complete timestamps
  - ↳ *WMS scan events*
- ✓ Shipment weight
  - `Weight_in_gms`
  - ↳ *WMS pack station*
- ✓ Warehouse zone
  - `Warehouse_block`
  - ↳ *WMS location master*

Primary system: WMS task and labor records

STEP 4 — Carrier Tender

DECISION

Expedite, switch carrier, or warn the customer?

QUESTIONS

- Will the carrier accept the load on time?
- Is this lane running late?

VARIABLES

- ◐ Carrier tender acceptance rate
  - ↳ *TMS tender history*
- ◐ Lane on-time rate, last 30 days
  - ↳ *Carrier scorecard spreadsheet*
- ◐ Hours from tender to required pickup
  - ↳ *TMS appointments*
- ✓ Shipment mode
  - `Mode_of_Shipment`
  - ↳ *Order header*

Primary system: TMS tender and scorecard

STEP 5 — Pickup & Transit

DECISION

Intervene while in transit?

QUESTIONS

- Did pickup happen on time?
- Is the ETA slipping?

VARIABLES

- ◐ Pickup delay vs. appointment
  - ↳ *Dock appointment log*
- ✗ Milestone ETA variance
  - ↳ *Carrier portal or EDI 214*
- ✗ Weather and congestion on route
  - ↳ *External data feed*

Primary system: carrier milestones, visibility platform

OUTCOME (Y) — Delivered On Time?

TARGET

Shipment late (1) or on time (0)

VARIABLE

- ✓ Target
  - `Reached.on.Time_Y.N`
  - ↳ *Proof of delivery*

Likely recorded during or after delivery. Exclude as predictors.

- ! Customer rating
  - `Customer_rating`
  - ↳ *Post-delivery customer survey*
- ! Care calls
  - `Customer_care_calls`
  - ↳ *Call center log*

Not mapped to any decision: `Prior_purchases`, `Gender`

✓ In the example dataset ◐ Captured elsewhere in the organization, missing from dataset

✗ Not captured today ! Timing violation (post-outcome) ↳ *Where the data stream lives*

▲ Wait / queue where lateness accumulates

6 of 22
predictor variables in the dataset

0 of 10
timing and queue variables

2
post-outcome fields to exclude